



The Federal University of Technology, Akure.
School of Engineering and Engineering Technology
Department of Information & Communication Technology
First Semester Examinations 2023/2024 Session

ICT 507: Electrical and ICT Installation Design

Date: May, 2024.

Instructions: Answer **THREE** questions in all, **ONE QUESTION** from section A and any other **TWO** questions from section B

Course Unit: 3

Time Allowed: 2 Hours

SECTION A

Question 1

- a. Justify the need for standards and regulations in electrical service design generally and state one specific role in each of the following regulatory bodies (i) IEC (ii) BS (iii) IEE (iv) NERC
(10 marks)

- b. An office has a length of 20m, width of 10m and height of 3m. Ceiling to desk height is 2m. area to be illuminated to a general is 250 lux using twin lamp 32 watt CFL Luminaire with a space to height ratio (SHR) of 1.25. Each lamp has an initial output of 85 lumen per watt. Maintenance factor is 0.63 and Utilization factor is 0.69. If the required lux level of the office is assumed to be 250, calculate:

- Total wattage of the luminaire
- The room index ratio
- Number of luminaire required
- Minimum spacing ratio between luminaire
- Number of row required
- Number of luminaire in each row
- Axial spacing between each fixture
- Transverse spacing between each fixture

(20 marks)

Question 2

- a. List and explain five basic electrical safety rules in accordance with electrical installation safety standards
(10 marks)
- b. You have been provided with the isometric view of a room with wiring installation plan as shown in Figure 2 with the following assumptions: Height of batten is 2.5m above the floor, Height of the switch board is 1.3m above the floor level, Height of the distribution board is 1.5m above the floor level. Heights of the brackets for light points is 2.5m above the floor level. The main switch is rated 250V, 15A DPIC and Distribution box is rated as 250V, 15A (two way iron clad). Take the wattages of the lighting points as 60watts, fan point as 100 watts and power socket 100 watt:

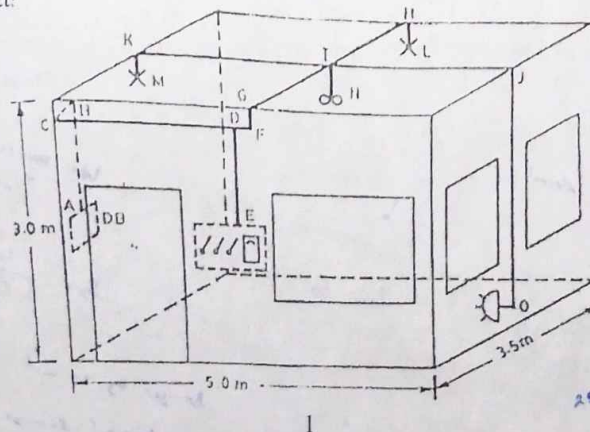


Figure 2

Equipment must be properly grounded and should be checked for proper use.

- i. Sketch the wiring installation plan of the room
- ii. Sketch the wiring diagram for the room
- iii. Determine the total no of sub circuit
- iv. Determine the size of cables needed for the installation
- v. Calculate the total length of wire required for the installation (phase – wire for each sub circuit and neutral wire)

(20 marks)

SECTION B

Question 3

- a. With a neatly drawn one line diagram, list all the components of Nigerian Power System.
(5 marks)
- b. What is/are the difference(s) between a wire and a cable? Distinguish between a cable insulation and cable mechanical protection
(5 marks)
- c. A cable with voltage drop figure of 6.4 mV/A/m supplies a current of 24 A to a point 18 m away from a 230 V supply source. Determine:
 - i. The voltage drop in the cable
 - ii. The actual voltage at the load point

(5 marks)

Question 4

- a. List five factors necessary in determining the choice of cable in electrical service design
(5 marks)
- b. Explain five steps involved in troubleshooting electrical problems
(5 marks)
- c. There is voltage drop limitation of 5 V for a circuit wired in PVC insulated twin and earth cable (clipped direct) having a length of run of 35 m . The current demand is assessed as 36 A and protection is by a BS 88 fuse. Establish the following:
 - i. Fuse rating,
 - ii. Maximum mV/A/m
 - iii. Minimum cable cross sectional area

(5 marks)

Question 5

- a. Write short notes on the following:
 - i. Insulation test
 - ii. Continuity test
 - iii. Earth leakage test
- b. The voltage drop for a certain cable is 44 mV/A/m . Calculate the voltage drop in a 15 m run of this cable when carrying a load of 6 A .
(5 marks)
- c. The design current of a circuit protected by a BS 88 fuse is 28 A , the grouping correction factor is 0.8 and the ambient temperature correction factor is 1.04 . Calculate the minimum current carrying capacity of the cable.
(5 marks)

(5 marks)

(5 marks)



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ELECTRICAL SYSTEMS ENGINEERING
DEPARTMENT OF INFORMATION & COMMUNICATION ENGINEERING
B. ENG. DEGREE EXAMINATIONS. FIRST SEMESTER 2024/2025 SESSION

COURSE CODE: ICT507 **COURSE TITLE:** Electrical and ICT Service Design

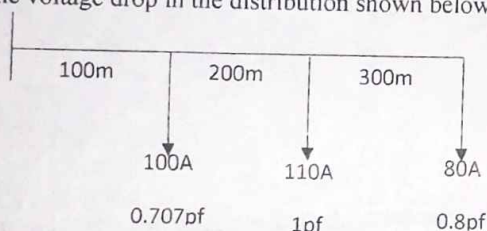
UNITS: 2.0

TIME ALLOWED: 2 Hours

Instruction: Answer **THREE** questions in all, **QUESTION ONE** and any other **TWO** questions

25/60

- ✓ 1a. Write short notes on the roles and importance the following regulatory bodies in Electrical service design and Installation (1.) IEC, (2.) BS, (3.) IEE, and (4) NERC (6 marks)
- 1b. Discuss six (6) Electrical safety rules in accordance with the Personal Protective Equipment (PPE) Regulations 2002 and the Personal Protective Equipment at Work Regulations 1992 (as amended). (6 marks)
- 1c. List six (6) factors to be considered in determining choice of cable in electrical service design (6 marks) ✓
- 1d. What are the step by step procedures to be followed in selection of cable sizes for an installation? (6 marks)
- 1e. Assuming that the resistance and reactance of copper in a loop of 1000m is 0.25 and 0.10 respectively, find the voltage drop in the distribution shown below.



(6 marks)

$V_d = \frac{W \times L \times R}{1000}$

20 marks

45/60

- ✓ 2a. Justify the need for correction factor (Cf) in cable selection (5 marks) ✓
- 2b. Write short notes on the following correction factors: (i) C_d , (ii) C_g , (iii) C_r , and (iv) C_i (5 marks) ✓
- 2c. Determine the main cable size PVC twin to supply 75.55A to a consumer unit of length 56m from the meter panel at 30°C. Let selected rating be 80A, Correction factors @ 30°C be 1 and in an enclosed system. Use the following IEE table values $80A \Rightarrow \begin{cases} 79 \Rightarrow 25mm^2 \Rightarrow 1.8mV v_d \\ 98 \Rightarrow 35mm^2 \Rightarrow 1.3mV v_d \end{cases}$ (5 marks) ✓
- ✓ 3a. With a neatly drawn one line diagram, list all the components of Nigerian Power System. (5 marks) ✓
- 3b. What is/are the difference(s) between a wire and a cable? Distinguish between a cable insulation and cable mechanical protection (5 marks)
- 3c. A cable with voltage drop figure of 6.4mV/A/m supplies a current of 24A to a point 18m away from a 230V supply source. Determine:
- a. The voltage drop in the cable
- b. The actual voltage at the load point (5 marks) ✓

12.5/15

11-25
11-330
330-132
132-33
33-11
11-0.46

Any xtr
-sec
-P
-sec
-ter



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- 4a. Explain five steps involved in troubleshooting electrical problems (5 marks) ✓
- 4b. Prepare a Bill of Engineering Measurement and Evaluation (BEME) for a proposed two bedroom residential building with all standard assumptions and your chosen material unit prices. (5 marks)

4c. There is voltage drop limitation of 5V for a circuit wired in PVC insulated twin and earth cable (clipped direct) having a length of run of 35m. The current demand is assessed as 36A and protection is by a BS 88 fuse. Establish the following:

- a. Fuse rating, ✓
- b. Maximum mV/A/m
- c. Minimum cable cross sectional area (5 marks) ✓

5a. Write short notes on the following:

- a. Insulation test
- b. Continuity test
- c. Earth leakage test (5 marks) ✓

5b. The voltage drop figure for a certain cable is 44mV/A/m. Calculate the voltage drop in a 15m run of this cable when carrying a load of 6A. (5 marks)

5c. The design current of a circuit protected by a BC 88 fuse is 28A, the grouping correction factor is 0.8 and the ambient temperature correction factor is 1.04. Calculate the minimum current carrying capacity of the cable. (5 marks). ✓

Insulation resistance test: A factor of 0.925 is used for the

Visual inspection ✓
Earth continuity test ✓
Insulation resistance test ✓
Residual Current Device (RCD) ✓

C_a - 100%

C_g - group - heat

C_f - A factor of 0.925 is used for the grouping factor.

C_i - A factor of 0.5 is used when the insulation resistance is less than 0.5 MΩ of the cable.